

SUMMARY

Machine Learning Engineer II at Amazon building AI, ML, Agent and data platforms in Python, Java, TypeScript, and AWS. Work covers the model and agent lifecycle: data pipelines, training, evaluation, deployment, monitoring, and production guardrails. Most relevant work includes agent platforms, AI/ML evaluation frameworks, SageMaker training pipelines, Spark-backed data access, and tooling for science and business teams.

SKILLS

- **Programming:** Python, Java, TypeScript, SQL
- **AI/ML Platforms:** Agentic Systems, MCP, Agent Evaluation, LLM-as-Judge, MLOps, Causal Inference, EconML
- **Data & Infrastructure:** Spark, AWS CDK, SageMaker, Bedrock, Step Functions, Glue, EMR, Redshift, Athena, Docker, CI/CD
- **Systems:** Observability, Access Control, Data Quality, A/B Experimentation, Search Systems, Enterprise Integration
- **Languages:** English, Japanese (JLPT N1), Mandarin Chinese

EXPERIENCE

- **Amazon (Japan Consumer Innovation)** Tokyo, Japan
Machine Learning Engineer July 2025 – Present
 - Architected and now operate an AI-powered analytics platform that lets retail teams query business metrics, knowledge, workflows, and science model outputs through natural language; built the online query path, offline ETL path, domain skill warehouse, and production observability.
 - Designed an agent evaluation framework with deterministic graders, SQL and tool-call validation, partial-credit result checks, and LLM-as-Judge scoring; made it the release gate for new agent capabilities and regression protection.
 - Built a causal inference ML framework on AWS and SageMaker, covering feature-store construction, training and validation data preparation, estimator comparison with EconML, and production training pipelines; reduced model iteration from weeks to hours.
 - Built a reusable AWS CDK data-access platform that lets teams connect project-owned AWS accounts to Amazon’s internal data warehouse for scheduled pipelines and ad-hoc analysis, replacing bespoke setup with standardized Spark, Redshift, Glue, EMR, IAM, and Lake Formation provisioning.
 - Mentor junior engineers and business intelligence engineers through project planning, technical design, implementation reviews, and career development.
- **Amazon (Product Search)** Tokyo, Japan
Software Development Engineer Apr 2023 – June 2025
 - Designed and built an offline evaluation system for search refinement selection, using traffic replay, feature flags, paired statistical analysis, relevance scoring, duplication analysis, and continuous monitoring to give scientists and engineers rapid pre-A/B quality signals.
 - Contributed to the launch and worldwide expansion of a Lucene-based refinement selection backend replacing a legacy navigation system; owned pre-launch evaluation, launch readiness, change management, and validation across regional rollout phases.

EDUCATION

- **Tokyo Institute of Technology** Tokyo, Japan
M.Eng. in Information and Communication Engineering Sep 2020 – Sep 2022
- **Huazhong University of Science and Technology** Wuhan, China
B.Eng. in Electronic Information Engineering Sep 2016 – Jun 2020